

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

**MOBILE APPLICATION FRAMEWORK FOR ENHANCING
CUSTOMER SERVICE IN INTERNET SERVICE PROVIDER**

BY
MA EAJINT CHAW SWE
VI IT - 8

GRADUATION THESIS

OCTOBER, 2025
THANLYIN

LIBRARY OF TTU

Thesis Accession No. BT - 1637

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

**MOBILE APPLICATION FRAMEWORK FOR ENHANCING
CUSTOMER SERVICE IN INTERNET SERVICE PROVIDER**

TECHNOLOGICAL UNIVERSITY (THANLYIN)

★ - 1 DEC 2025 ★

LIBRARY

BY

MA EAIN T CHAW SWE

VI IT - 8

REFERENCE

GRADUATION THESIS

OCTOBER, 2025

THANLYIN

Technological University (Thanlyin) Library

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

**MOBILE APPLICATION FRAMEWORK FOR ENHANCING
CUSTOMER SERVICE IN INTERNET SERVICE PROVIDER**

BY
MA EAIN T CHAW SWE
VI IT - 8

A THESIS
SUBMITTED TO THE DEPARTMENT OF
INFORMATION TECHNOLOGY ENGINEERING
IN PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
(INFORMATION TECHNOLOGY ENGINEERING)

OCTOBER, 2025
THANLYIN

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

We certify that we have examined, and recommend to the University Steering Committee for Bachelor of Engineering Graduation thesis entitled: “**MOBILE APPLICATION FRAMEWORK FOR ENHANCING CUSTOMER SERVICE IN INTERNET SERVICE PROVIDER**” submitted by **MA EAJNT CHAW SWE**, Roll No. **VI IT-8 (NOVEMBER, 2024)** to the Department of Information Technology Engineering as the requirement for the Bachelor of Engineering degree, **BE(IT)**.

Board of Examiners:

1. Dr. Mon Mon Zaw

Professor and Head

Department of Information Technology Engineering
Technological University (Thanlyin)

.....
(Chairman and Supervisor)

2. Daw Nge

Associate Professor

Department of Information Technology Engineering
Technological University (Thanlyin)

.....
(Member)

3. Daw Hay Mun Oo

Lecturer

Department of Information Technology Engineering
Technological University (Thanlyin)

.....
(Member)

4. Daw Swe Thiri Maw

Demonstrator

Department of Information Technology Engineering
Technological University (Thanlyin)

.....
(Member)

5. Daw Wyne So

Demonstrator

Department of Information Technology Engineering
Technological University (Thanlyin)

.....
(Member)

ACKNOWLEDGEMENT

The author would like to express my deepest gratitude to Dr. Theingi, Rector of Technological University (Thanlyin), for granting permission to undertake this thesis.

Sincere appreciation is extended to Dr. Mon Mon Zaw, Professor and Head of Department of Information Technology Engineering, Technological University (Thanlyin), for her guidance, thoughtful corrections, and valuable suggestions throughout this work. As the supervisor, her kind support and constructive advice were instrumental to the successful completion of this research.

The author would also like to express thanks to the Board of Examiners from the Department of Information Technology Engineering for their insightful comments, generous time, and willingness to provide guidance during the evaluation of this thesis.

Special thanks go to my parents, whose unwavering love, encouragement, and continuous support made it possible to complete this work without difficulty.

ABSTRACT

The increasing reliance on digital service platforms has created a strong demand for efficient, scalable, and user-friendly tools to manage both administrative operations and customer interactions. For Internet Service Providers (ISPs), this demand is particularly significant, as providers must efficiently handle large volumes of user data, manage subscription lifecycles, process complaints, and deliver reliable support. Manual approaches are often inefficient, whereas modern application-based solutions improve automation, transparency, and service quality.

This thesis focuses on the development and evaluation of two interconnected applications designed for ISP operations: an administrator application and a customer application. The administrator application centralizes control over core processes, including user record management, subscription approvals, payment verification, complaint handling, and support chat. It also provides tools for managing township coverage and bank details while monitoring device and service status. These functionalities ensure smooth service delivery and efficient oversight.

The customer application emphasizes accessibility and ease of use. It enables customers to register and log in, manage personal profiles, explore and purchase internet packages, and upload payment proofs for verification. Users can track subscription statuses, submit complaints, and interact with support teams in real time. This reduces administrative burden while enhancing customer transparency and engagement.

Beyond application development, this research explores the integration of data-driven approaches to improve ISP service quality. By identifying operational data from the applications, machine learning techniques could be applied for real-time alerts, automated complaint prioritization, and adaptive service recommendations. Such features would further optimize ISP operations while fostering greater customer satisfaction and retention.

TABLE OF CONTENTS

	Page
ACKNOWLEDGEMENT	i
ABSTRACT	ii
TABLE OF CONTENTS	iii
LIST OF FIGURES	v
 CHAPTER TITLE	
1 INTRODUCTION	1
1.1. Introduction to ISP Management and Customer Application	1
1.2. Problem Statement	1
1.3. Aim and Objectives of Thesis	2
1.4. Scope of Thesis	2
1.5. Outlines of Thesis	3
2 BACKGROUND THEORY	4
2.1. Background Theory	4
2.2. Technologies and Tools	4
2.2.1. Cloud Computing	4
2.2.2. Serverless Architecture	5
2.2.3. Flutter & Dart	5
2.2.4. Supabase Platform	6
2.2.5. PostgreSQL as the Core Database in Supabase	7
2.2.6. Development Tools: Android Studio and VSCode	7
2.3. Summary	8
3 SYSTEM DESIGN OF THE SYSTEM	9
3.1. System Design and Methodology of the Internet Service Provider Applications	9
3.1.1. Overview of ISP's Customer Service Department	9
3.2. System Design for Administrator Panel	10
3.3. System Design for User Panel	12
3.4. Algorithms	14
3.4.1. Unique Identifier Generation	14
3.4.2. Time and Date Arithmetic	15
3.4.3. Periodic Polling and Auto-Refresh	15

	3.4.4. Data Grouping and Aggregation	16
	3.4.5. Layout Geometry and Chart Computation	17
	3.4.6. Text Filtering and Search	17
	3.4.7. SMTP and MIME Encoding	18
	3.5. Summary	19
4	IMPLEMENTATION OF THE SYSTEM	20
	4.1. Implementation of Mobile Application Framework	20
	4.2. System Operations by Administrator	19
	4.2.1. Admin Panel Dashboard	21
	4.2.2. Admin Dashboard Showing User Activity and Inventory Overview	21
	4.2.3. Customer List Operation	23
	4.2.4. Managing Administrator Section	24
	4.2.5. Operating Banner	24
	4.2.6. Managing Packages Section	26
	4.2.7. Adding Bank Information	28
	4.2.8. Managing Townships' Information	29
	4.2.9. Comprehensive View of All Registered Subscriptions	31
	4.2.10. Configuring Server and Device IP Addresses for New Users	33
	4.2.11. Device Connectivity Indicators in the Network Simulation Tool	34
	4.2.12. Enhancing Customer Service via the Chat Section	37
	4.2.13. Managing Complaints Section	38
	4.3. System Operations in User Application	39
	4.3.1. User Registration	40
	4.3.2. The Dashboard View of Subscription History	46
	4.3.3. Making Ticketing via ISP User Application	47
	4.3.4. Real-Time Chat Feature in the ISP User Application	48
	4.4. Summary	49
5	DISCUSSION AND CONCLUSION	50
	5.1. Discussion	50
	5.2. Conclusion	51
	REFEREN	
	CES	52